Machine-Checked Dimension Theory for Fusion Categories

A research proposal: pivotal and spherical structures, the double-dual twist, and a verified proof of Etingof–Nikshych–Ostrik Proposition 8.23

Tzu-Chen Huang

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Abstract

This proposal sets out a research programme whose central target is the first machine-checked proof of Etingof–Nikshych–Ostrik Proposition 8.23: *every pseudo-unitary fusion category admits a unique spherical structure whose dimensions are the Frobenius–Perron dimensions.* The theorem is the standard bridge from an analytic hypothesis (one scalar equation on the global dimension) to the algebraic structure that all of dimension theory presupposes, and it is used as a black box throughout the classification literature. No formal proof of it exists, and the categorical infrastructure on which it rests is only now arriving in Mathlib.

The programme builds on substantial preliminary work, reported in §4: a skeletal formalisation of multiplicity-free rigid fusion data with a proved inhabitant; a corrected duality normalisation that inhabits Fibonacci and Ising over $\mathbb{R}$; a reduction of the target to a single per-channel identity on the double-dual twist; a formalised proof of ENO Proposition 8.21; and a reconstruction of ENO's own route through the square root a of the quadruple-dual isomorphism. Two of these results are negative and change the shape of the target: ENO 8.23 is *false* without its pseudo-unitarity hypothesis (a kernel-checked two-simple counterexample, the Yang–Lee category), and the number and identity of pivotal structures is not a function of the fusion ring (a kernel-checked twisted $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$ model). The proposal turns these findings into a three-aim programme: a faithful and inhabitable formal signature together with its bridge to Mathlib's emerging pivotal and spherical category theory; a derivation of the double-dual twist from the duality data and a proof of the multiplicativity identity that ENO's argument identifies with the trace identity; and a formalisation of ENO's analytic core, yielding Proposition 8.23 with positivity and uniqueness exactly as stated. Deliverables include an axiom-clean Lean 4 library, a verified negative-result note, upstream Mathlib contributions, and a documented methodology for auditing formal statements against their sources.

At a glance.

Object	Pivotal and spherical structures on fusion categories, and ENO Prop. 8.23.
Method	Skeletal finite data in Lean 4 (pinned to Mathlib `0df444a3`); exact arithmetic in $\mathbb{Z}[\sqrt{d}]$; kernel-checked counterexamples; axiom audit; bridge to Mathlib's categorical layer.
Already done	∼5,200 lines, 23 modules, 3,032-job green build, 73 audited declarations at `[propext, Classical.choice, Quot.sound]`, no `sorry`.
Still open	Derivation of the twist J and the identity (M); the positivity step; the general (non-self-dual, non-multiplicity-free) signature; the categorical bridge.
Horizon	24 months, 7 work packages, 12 numbered deliverables.

Contents

1 Introduction **3**
1.1 Dimension theory sits between decidable combinatorics and open structure . . . 3
1.2 The theorem that bridges the two . . . 3
1.3 Why formalise, and why this is not transcription . . . 3
1.4 What has already been established . . . 4
1.5 What is not yet established . . . 4
1.6 Research questions . . . 4
1.7 Structure of the proposal . . . 5

2 Background **5**
2.1 Fusion categories and fusion rings . . . 5
2.2 The Frobenius–Perron dimension . . . 6
2.3 Duality, pivotal structures, and the double-dual twist . . . 6
2.4 Pseudo-unitarity and the target . . . 7
2.5 The skeletal point of view and what it costs . . . 7

3 Prior work **7**
3.1 The mathematical sources . . . 7
3.2 The formalisation landscape . . . 9
3.3 The author's prior formalisation . . . 9

4 Preliminary results **10**
4.1 The formal signature . . . 10
4.2 Four defects, and what caught them . . . 10
4.3 The duality normalisation . . . 11
4.4 The double-dual twist, and the two identities . . . 11
4.5 Refutations . . . 12
4.6 ENO Proposition 8.21 . . . 13
4.7 ENO's own route: the square root a and the b-cochain . . . 13
4.8 The instances, and the audit . . . 14
4.9 The exact gap . . . 14

5 Research programme **14**
5.1 Three aims, and the logical geography of the target . . . 14
5.2 Work packages . . . 15
5.3 Milestones . . . 19

6 Techniques **19**
6.1 Finite data and exact arithmetic . . . 19
6.2 Inhabitant-first statement design . . . 20
6.3 Falsification before proof . . . 20
6.4 Multiplicities: why this is a separate development . . . 20
6.5 Diagram evaluation and independent reference computations . . . 21
6.6 Parallel adversarial exploration . . . 21
6.7 Statement fidelity as a method . . . 22
6.8 Engineering discipline . . . 22

7 Risks and mitigations **22**

8 Timeline **23**

9 Expected deliverables **23**

10 Significance **25**

A Module inventory and dependency graph **27**

B The audit protocol **28**

C The main statements **28**

1 Introduction

1.1 Dimension theory sits between decidable combinatorics and open structure

A fusion category over a field $\Bbbk$ is a $\Bbbk$-linear, semisimple, rigid monoidal category with finitely many simple objects and $\mathrm{End}(\mathbf{1}) = \Bbbk$. Its Grothendieck ring $K_0(\mathcal{C})$ is finite, decidable data: a finite set I of simples, integers $N_{ij}^k = \dim \mathrm{Hom}(X_k, X_i \otimes X_j)$, a duality involution $i \mapsto i^*$. Almost everything one wants to compute about $\mathcal{C}$ — dimensions, the global dimension, Frobenius–Schur indicators, the T-matrix of a modular category, the state sum of a three-manifold — depends on more than this ring, but less than the full category.

The intermediate datum is a *pivotal structure*: a monoidal isomorphism $\gamma\colon \mathrm{Id} \to (-)^{**}$. It yields a dimension $\dim_\gamma(X) = \mathrm{tr}(\gamma_X)$ for every object, and a *spherical* structure — one with $\dim_\gamma(X) = \dim_\gamma(X^*)$ — is what makes the trace cyclic and therefore produces invariants of oriented manifolds rather than of framed ones.

Two facts locate the research problem precisely.

(a) *The dimension function is not a function of the fusion ring.* This is not a philosophical remark: the present project contains a kernel-checked witness (§4, `VecZ2Twisted`). The category $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$ with non-trivial cocycle carries two pivotal structures whose dimensions on the non-trivial simple are $+1$ and -1 respectively. Any formalisation that treats "the dimension" as determined by the fusion rules is therefore wrong, and the first version of this project's pivotal axiom was wrong in exactly that way.

(b) *Whether a pivotal structure exists is open in general.* Etingof–Nikshych–Ostrik conjectured that every fusion category is pivotal (ENO, Conjecture 2.8); the spherical strengthening is stronger still. What is known is a sufficient condition of a different character: pseudo-unitarity.

1.2 The theorem that bridges the two

ENO Proposition 8.23 states that a pseudo-unitary fusion category — one whose categorical global dimension equals its Frobenius–Perron global dimension — admits a *unique* pivotal structure, which is spherical, and with respect to which the categorical dimensions of all simple objects are positive and equal to their Frobenius–Perron dimensions.

The proposition is used as a black box across the classification programme: in Ostrik's analysis of pivotal fusion categories of rank 3 and the resulting inequality, in the Haagerup–Izumi existence problem, in the passage from unitary anyon data to modular data, and in every place where one writes down a spherical category from a fusion ring plus positivity. It is, in a precise sense, the lemma that makes the rest of the subject work.

1.3 Why formalise, and why this is not transcription

A referee reading a two-page proof in a 2005 Annals paper cannot check, line by line, that the gauge chosen for the associator is the one in which the evaluation and coevaluation have the normalisation the argument assumes. A proof assistant can. The empirical case for this proposal is that when the present project first wrote down the statement, four defects surfaced, every one of them invisible to a reader checking that the formulas *looked* right:

D1. The signature had no inhabitants. The two unit axioms of the skeletal fusion datum were mutually inconsistent for $n \geq 2$: evaluated at $(0, b, 0, d, b, 0)$ they forced $1 = 0$. Every theorem quantified over the structure was therefore vacuous, including a "proof" of the target theorem.

D2. The pentagon was not the pentagon. The axiom called `pentagon` had eight free labels where the F-matrix pentagon requires ten, and it fails on $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$ for *every* cocycle ω, the trivial one included. A green build certified a false axiom.

D3. The pivotal axiom fixed the twist to one. `p_fusion` assumed $J \equiv 1$, where J is the monoidal structure of $(-)^{**}$. With that assumption every fusion-rule table is pivotal via $p \equiv 1$, and the target theorem became provable with no hypothesis at all — the formal statement had drifted so far from the source that it asserted something false.

D4. The second snake used the wrong inverse. It wrote the right snake with a reciprocal of an associator *entry* where the $(0,0)$-entry of the *matrix inverse* of the associator block is required. The two agree for pointed categories and differ for Fibonacci, so the axiom excluded the two principal examples.

Each defect was found by making the statement inhabitable and checking it against the examples, not by re-reading it. This is the methodological thesis of the proposal: in this subject the value of formalisation lies less in the final certificate than in the audit that produces it, and the audit is transferable.

1.4 What has already been established

The preliminary results of §4 are not a plan; they are a library. In summary:

- a skeletal signature for multiplicity-free rigid fusion data, with a proved inhabitant ($\mathrm{Vec}_{\mathbb{Z}/2}$) and the pentagon checked on all 512 label tuples;
- a corrected duality normalisation, with the inverse associator carried as data, that inhabits Fibonacci and Ising over $\mathbb{R}$ — every field proved, including both snakes — and yields the spherical pivotal structure with dim = FPdim for both;
- a proof that a pivotal dimension is always a coboundary of the normalised double-dual twist, and that ENO 8.23 is *equivalent* to one per-channel identity on that twist;
- a formalised ENO Proposition 8.21: the spectral bound $|d_i| \le \mathrm{FPdim}_i$ for any dimension function satisfying the eigenvector equation, together with the equality case;
- a kernel-checked refutation of the target as stated without hypotheses (Yang–Lee), and a second witness that pivotal structures are not determined by the fusion ring;
- a reconstruction of ENO's own proof route — the square root a of the quadruple-dual isomorphism, its monoidality defect b, and the trace identity — together with a proof that ENO's trace identity *is* the multiplicativity identity that the twist reduction isolates.

The library builds (3,032 jobs, warnings only), contains no `sorry`, and its axiom audit over 73 declarations reports `[propext, Classical.choice, Quot.sound]` throughout.

1.5 What is not yet established

The gap is now sharp, and it is not the algebra. It is the passage from a Hermitian structure on a pseudo-unitary category to positivity and multiplicativity of a chosen dimension function — ENO's analytic argument: Proposition 8.21 turns the *sum* equality $\sum_i |d_i|^2 = \sum_i \mathrm{FPdim}_i^2$ into a termwise equality $|d_i| = \mathrm{FPdim}_i$, and two tensor automorphisms ($\sigma = d^2/|d|^2$ and $\tau = d/|d|$) then produce the pivotal structure with positive dimensions. Alongside it, and logically prior to it, sits a defect of the signature rather than of the mathematics: the twist J is carried as data constrained by four laws, not derived from $(\mathrm{ev}, \mathrm{coev}, F)$. Until it is derived, the statement "p is pivotal" mentions an object the mathematics does not fix.

1.6 Research questions

Q1 (Signature).
What is the weakest finite-data signature that is (i) inhabited by the standard examples, (ii) strong enough to express pivotality and sphericity faithfully, and (iii) provably equivalent to the abstract categorical notion?

Q2 (Twist).
Is the double-dual twist J determined by $(F, \mathrm{ev}, \mathrm{coev})$? If so, what is its closed form, and does it satisfy the multiplicativity identity (M)?

Q3 (Analytic core).
Can ENO's proof of Proposition 8.23 be carried out inside the skeleton, taking as input only the pseudo-unitarity equation $\dim \mathcal{C} = \mathrm{FPdim}\mathcal{C}$?

Q4 (Frontier).
What is the largest statement provable *without* pseudo-unitarity — that is, what exactly is the content of ENO Conjecture 2.8 in a pivotal-free formalism, and which of its consequences survive as theorems about torsion in the universal grading group?

Q5 (Method).
Which failure modes of formalising monoidal category theory are systematic, and can they be detected mechanically rather than by luck?

1.7 Structure of the proposal

§2 fixes notation and states the target precisely. §3 surveys prior work, mathematical and formal. §4 reports the preliminary results in enough detail to be checked. §5 sets out the seven work packages and their milestones. §6 describes the techniques, including the ones that failed. §7–§9 give risks, timeline and deliverables. §10 argues for significance. Three appendices record the module inventory, the audit protocol, and the Lean statements of the main targets.

2 Background

2.1 Fusion categories and fusion rings

Definition 2.1. *A* fusion category *over a field* $\Bbbk$ *is a* $\Bbbk$*-linear abelian, semisimple, rigid monoidal category* $\mathcal{C}$ *with finitely many isomorphism classes of simple objects, such that* $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) \simeq \Bbbk$ *and the tensor product is bilinear and exact in each variable. We write* I *for the set of isomorphism classes of simples,* X_i *for a representative,* $0 \in I$ *for the class of* $\mathbf{1}$*, and* $i \mapsto i^*$ *for the duality involution determined by* $\mathrm{Hom}(X_i \otimes X_j, \mathbf{1}) \neq 0 \iff j = i^*$.

The Grothendieck ring $K_0(\mathcal{C})$ has $\mathbb{Z}$-basis (X_i) and structure constants

$$X_i X_j = \sum_{k \in I} N_{ij}^k X_k, \qquad N_{ij}^k = \dim_{\Bbbk} \mathrm{Hom}(X_k, X_i \otimes X_j) \in \mathbb{N}.$$

In this proposal a *fusion ring* is the resulting abstract datum $R = (N, (-)^*, 0)$: the multiplicities, the duality involution, and the unit. The axioms used are those of a based ring: $N_{ij}^0 = \delta_{j,i^*}$; associativity $\sum_k N_{ij}^k N_{kl}^m = \sum_k N_{jl}^k N_{ik}^m$; and compatibility of the involution with the product. Commutativity is *not* assumed.

Definition 2.2. *A fusion ring is* multiplicity-free *if* $N_{ij}^k \in \{0, 1\}$ *for all* i, j, k*. It is* pointed *if every simple object is invertible, i.e.* $N_{ij}^k \in \{0, 1\}$ *and* $\sum_k N_{ij}^k = 1$ *for all* i, j.

Multiplicity-freeness is not a convenience in this project: with $N > 1$ the associator is a matrix between multi-dimensional Hom spaces, and a single scalar $F_d^{abc}(e, f)$ cannot hold it. The restriction is discussed in §6.4.

2.2 The Frobenius–Perron dimension

Theorem 2.3 (Frobenius–Perron, for fusion rings)**.** *Let R be a fusion ring with n classes. There is a unique function $f\colon I \to \mathbb{R}_{>0}$ with $f_0 = 1$ and*

$$f_i f_j = \sum_k N_{ij}^k f_k \qquad \text{for all } i, j \in I. \tag{1}$$

Moreover $f_{i^*} = f_i$, each f_i is an algebraic integer, and if the fusion graph is connected then f is the Perron–Frobenius eigenvector of every fusion matrix.

The number f_i is the *Frobenius–Perron dimension* $\mathrm{FPdim}(X_i)$, and $\mathrm{FPdim}(\mathcal{C})^2 = \sum_i f_i^2$ is the Frobenius–Perron global dimension. Note that (1) is the *character* form: f is a ring homomorphism $K_0(\mathcal{C}) \to \mathbb{R}$.

In the present library Theorem 2.3 is not proved: the existence and uniqueness of f is packaged as a structure `FPData`, supplied by hand for each example. Discharging it from Mathlib's Perron–Frobenius development is a deliverable (§5, WP0), not a research risk: the relevant material exists upstream as a chain of pull requests (§3).

2.3 Duality, pivotal structures, and the double-dual twist

Rigidity supplies, for each object X, an evaluation $\mathrm{ev}_X\colon X^* \otimes X \to \mathbf{1}$ and a coevaluation $\mathrm{coev}_X\colon \mathbf{1} \to X \otimes X^*$ satisfying the two snake equations. In a skeleton, both are scalars, and the composite

$$\kappa_i \;=\; \mathrm{ev}_i \cdot \mathrm{coev}_i \;\in\; \Bbbk^\times$$

is the loop scalar of X_i. It is *not* the dimension: the dimension depends on a choice of identification of X with its double dual.

Definition 2.4. *A* pivotal structure *on $\mathcal{C}$ is a monoidal natural isomorphism $\gamma\colon \mathrm{Id}_{\mathcal{C}} \to (-)^{**}$. Its* dimension *is $\dim_\gamma(X) = \mathrm{tr}(\gamma_X\colon X \to X^{**})$, computed as the composite of γ_X with the evaluation of the right dual. The structure is* spherical *if $\dim_\gamma(X) = \dim_\gamma(X^*)$ for every X.*

Two remarks are essential and both cost the project a rewrite.

Remark 2.5 (The twist). The functor $(-)^{**}$ is monoidal, and its monoidal structure is *not* the identity. Write J_{ij}^k for the scalar of $X_i^{**} \otimes X_j^{**} \to (X_i \otimes X_j)^{**}$ on the channel k. Then a pivotal structure on a skeleton is exactly a family $(p_i)_{i\in I} \subset \Bbbk^\times$ with

$$p_0 = 1, \qquad p_k = J_{ij}^k\, p_i\, p_j \quad \text{whenever } N_{ij}^k > 0, \tag{2}$$

and the resulting dimension is

$$\dim_i \;=\; p_i\, \kappa_i. \tag{3}$$

Sphericity becomes

$$\dim_{i^*} = \dim_i \;\; \forall i \qquad \Longleftrightarrow \qquad \kappa_{i^*} = p_i^2\, \kappa_i \;\; \forall i. \tag{4}$$

Remark 2.6 (The trap). It is tempting to assume that a skeleton can always be chosen with $(-)^{**} = \mathrm{id}$ on labels, so that (2) reads $p_k = p_i p_j$. That assumption is *precisely* pivotality: identifying the left and the right dual of an object is the content of a pivotal structure, and it is what need not exist. The project adopted this claim once, on the strength of a plausible argument that X^* is both a left and a right dual of X; the argument is wrong, and the claim is retracted in `PLAN.md`. A signature that builds the assumption in cannot express the general case.

2.4 Pseudo-unitarity and the target

Definition 2.7 (ENO §8.4)**.** *A fusion category $\mathcal{C}$ is* pseudo-unitary *if* $\dim \mathcal{C} = \mathrm{FPdim}\mathcal{C}$*, where* $\dim \mathcal{C}$ *is the categorical global dimension computed with respect to any pivotal structure. In the skeleton this is an equation without a pivotal structure in it:*

$$\sum_{i \in I} \kappa_i \, \kappa_{i^*} \;=\; \sum_{i \in I} f_i^2. \tag{5}$$

The equality (5) is form-independent because $\dim_i \dim_{i^*} = \kappa_i \kappa_{i^*}$ for every pivotal structure; the library proves this as `dim_mul_dim_dual`.

Theorem 2.8 (ENO Proposition 8.23; EGNO Theorem 9.6.10)**.** *Let $\mathcal{C}$ be a pseudo-unitary fusion category. Then $\mathcal{C}$ admits a pivotal structure γ; γ is spherical;* $\dim_\gamma(X) > 0$ *for every $X \neq 0$;* $\dim_\gamma(X_i) = \mathrm{FPdim}(X_i)$ *for every simple X_i; and γ is the unique such structure.*

The two supporting results that ENO use are:

Theorem 2.9 (ENO Proposition 8.21)**.** *Let $d\colon I \to \mathbb{C}$ satisfy the eigenvector equation $\sum_j N_{ij}^k d_j = d_i d_k$ for all i, k. Then $|d_i| \leq \mathrm{FPdim}_i$ for all i. If moreover $\sum_i |d_i|^2 = \sum_i \mathrm{FPdim}_i^2$, then $|d_i| = \mathrm{FPdim}_i$ for all i.*

Theorem 2.10 (ENO Theorem 2.6, after Radford)**.** *For any fusion category there is a monoidal natural isomorphism $g\colon \mathrm{Id} \to (-)^{****}$.*

Theorem 2.10 is taken as a hypothesis in this project (§4): proving it requires the weak Hopf algebra machinery of ENO §4–5 and is explicitly out of scope. Conjecture 2.8 of ENO — that every fusion category admits a pivotal structure — is not attacked directly; WP4 in §5 explains what the programme can say about it.

2.5 The skeletal point of view and what it costs

The formalisation works with finite data over a field $\Bbbk$:

$$(N_{ij}^k,\ i^*,\ F_d^{abc}(e,f),\ F^{-1},\ \mathrm{ev}_i,\ \mathrm{coev}_i,\ J_{ij}^k)$$

subject to the pentagon, the unit normalisations, the two inverse laws for the associator, the two snake equations, and the twist laws. This is the "F-symbol" presentation familiar from the anyon-model literature, and it has three costs, each addressed by a work package.

(i) **No abstract coherence.** A skeletal datum is a category only after a reconstruction theorem. WP5 supplies the bridge to Mathlib's categorical definitions and deletes the vendored patch once the upstream pull requests land.

(ii) **No labels with $i^{**} \neq i$.** The type of J has no slot for a label transport, and `p_mon` reads $p_k = J_{ij}^k p_i p_j$ with k the channel of $i \otimes j$. Every model in the library has $i^{**} = i$. WP1 repairs the signature; until then the F-free coherence law for J is the correct law only in that case.

(iii) **Multiplicity-freeness.** WP1 also relaxes $N \leq 1$; §6.4 explains why this is a separate development rather than a tweak.

3 Prior work

3.1 The mathematical sources

The source theorem. The target is Etingof–Nikshych–Ostrik, *On fusion categories*, Annals of Mathematics **162** (2005), 581–642 [ENO]. The relevant material is §2.2 (the quadruple dual), §3 (the T-matrix and the bound), and §8.4 (pseudo-unitarity and the spherical structure). Two

numbering conventions circulate: the published Annals version states the result as Proposition 8.23, while the arXiv version (`math/0203060v2`) states it as Proposition 8.22, one lower throughout §8.4. This proposal follows the Annals numbering and the library cites the number on every use; mis-numbering this proposition has already caused one wrong citation in the project's own plan.

The textbook treatment. EGNO, *Tensor Categories* [EGNO], Chapter 9 states the theorem as Theorem 9.6.10 with the corollaries 9.6.11–9.6.12, and is the more readable treatment. It is also the source of the convention that a pivotal structure is a monoidal isomorphism $\mathrm{Id} \to (-)^{**}$ rather than a choice of duals, which matters: the two formulations differ by a twist.

An independent, coordinate-free proof. Bartlett, *Fusion categories via string diagrams* [Bar15], gives a graphical development of exactly the results this project needs, and is the single most useful prior work for the proposed derivation of J:

- Theorem 3.28 and Corollary 3.30: the quadruple dual functor is monoidally isomorphic to the identity (ENO Theorem 2.6);
- Theorem 4.17: an *explicit formula* for the pivotal operators T^i_{jk} — the monoidal coherence of the canonical pivotal endofunctor — as the product of the *apex associator monodromy* and the *pivotal indicators*; this is the closed form the present project is missing for J;
- Theorem 5.4: pivotal structures on a (unitary) fusion category correspond exactly to tuples $(\gamma_i) \subset U(1)$ satisfying $\gamma_j\gamma_k = \gamma_i T^i_{jk}$ on every fusion channel, and sphericality is $\gamma_i = \pm 1$ for all i;
- Corollary 5.5: a pseudo-unitary fusion category admits a canonical spherical structure (the same statement as the target);
- §5.3: the *sphericalization* $\widetilde{\mathcal{C}}$ of $\mathcal{C}$ by the canonical $\mathbb{Z}/2$-action — the categorical form of the cover construction of WP4 in §5.

Theorem 5.4 is the cleanest available check on any derived formula for J: the two conventions are reciprocal (the present library's $p_k = J^k_{ij} p_i p_j$ against Bartlett's $\gamma_i = \gamma_j\gamma_k / T^i_{jk}$), and an index-level comparison gives an independent confirmation that does not go through the snake axioms at all. WP2 includes this comparison as an acceptance test.

Applications and the shape of the literature. Ostrik [Ost13] analyses pivotal fusion categories of rank three and derives the inequality

$$\sum_i d_i^2 \;\geq\; \sum_i \left|\textstyle\sum_j N^k_{ij} d_j\right|^2 / \ldots$$

that pins down the possible dimension functions; it is the place where "pivotal + positive + spherical" is used as a computational tool rather than as a structural hypothesis, and it is the template for the negative tests of WP6. The Haagerup–Izumi programme and the author's own work [Hua26] use the theorem at a different point, to convert coefficient identities (analytic) into a spherical structure (algebraic). Frobenius–Schur indicator theory [NS] is the other major consumer of the same layer: indicators are traces of pivotal structures composed with the duality, and their integrality is a sphericity statement.

What is *not* in the literature. (a) No proof of ENO Conjecture 2.8 (every fusion category is pivotal); the spherical version is strictly stronger and equally open. The counterexamples to naive expectations are of two kinds: categories with no pivotal structure at all are not known, but categories where the dimension depends on the pivotal structure are *abundant* (the present project has a kernel-checked one, §4). (b) No computation of the double-dual twist J in closed form for general multiplicity-free data; Bartlett's Theorem 4.17 is the closest, and it is stated through graphical monodromy rather than as an F-matrix formula ready for formalisation.

3.2 The formalisation landscape

Mathlib. `CategoryTheory.Monoidal.Rigid.Basic` provides `RigidCategory` (left and right rigid monoidal categories), with the file itself listing "define pivotal categories" as future work. There is no pivotal, spherical, or fusion category in Mathlib at the pinned revision, and no Perron–Frobenius theory for general irreducible non-negative matrices (only `Matrix.IsIrreducible`). The relevant upstream work is in flight:

- **PR #42145**, *the double right dual functor is monoidal* (+233/-4): defines `doubleRightDualFunctor` and its monoidal structure;
- **PR #42150**, *pivotal categories* (+370/-13 in 5 files): defines `PivotalCategory` as a rigid category with a monoidal natural isomorphism between the double dual functor and the identity; currently open and labelled *blocked-by-other-PR* on #42145;
- **PRs #42191** (spherical) and **#42192** (symmetric rigid is spherical), which build on #42150;
- the Perron–Frobenius chain **#39914**–**#39922**, whose declarations `PerronFrobenius.Irreducible`, `pft_irreducible`, `collatzWielandt` and `spectralRadius` supply the spectral theory the fusion-matrix argument needs.

The project's convention is to vendor a byte-compatible copy under `FusionCategory/MathlibPatches/` and to delete it, importing instead, when the pull requests merge. At the time of writing the vendored copies are stubs: `PivotalCategory.lean` is 33 lines and `PerronFrobenius.lean` is 22. This is honest debt, and WP0/WP5 repay it.

Other formal and semi-formal resources. Alphabeta Math's generated library page *Unitary Fusion Spaces, F/R Symbols, and Anyon Models* [AM] has multiplicity-aware fusion trees, the F-symbol as a recoupling matrix, the pentagon in fusion-tree coordinates, gauge covariance, and explicit Fibonacci and Ising carriers; it is pipeline-generated with provenance labels and is not a proof assistant development. TNLean [TN] is a blueprint project on tensor networks whose anyon chapter treats quantum-double models — a different object (microscopic lattice models rather than abstract fusion data). AnyonWiki [AW] supplies F/R/pivotal tables in a fixed gauge for multiplicity-free categories up to rank seven; it is *data*, not code, and the present project's Fibonacci and Ising conventions follow it rather than being derived afresh.

Conclusion. There is no formalisation of pivotal or spherical fusion categories anywhere, and the categorical infrastructure is landing in Mathlib during the period of this proposal. The strategic consequence is fixed: keep the skeletal layer aligned with #42150/#42191 naming, stub the bridge, and delete the patch when it merges. Reimplementing rigid category theory would be wasted effort; the value added here is the finite-data layer, the examples, the counterexamples, and the theorem.

3.3 The author's prior formalisation

The sibling package `haagerup-odd/` in the same repository is a completed formalisation of the main theorem of [Hua26]: `exists_EGEquations_odd`, the existence of the Evans–Gannon equations for the cyclic Haagerup–Izumi categories at every odd order. It comprises 115 modules and 24,295 lines, with 1,162 theorems and lemmas, no `sorry`, and an axiom audit reporting `[propext, Classical.choice, Quot.sound]`. It is pinned to the same toolchain as this project (Lean `v4.33.1`, Mathlib `0df444a3`).

Its relevance to this proposal is threefold. First, *feasibility*: the proposed library is roughly a fifth of that development in size, and the working method (finite data, exact arithmetic, per-file builds, an axiom audit in CI) is the one already proven at that scale. Second, *conventions*: both packages present fusion data as F-symbols with fixed gauges, and the Haagerup–Izumi coefficients are the natural large-scale instance for WP6. Third, *scope discipline*: that development stops at

Chapter 9 of the source manuscript and excludes the categorical reconstruction, exactly as this one excludes the weak Hopf algebra input of ENO §4–5.

4 Preliminary results

This section reports what is already proved, in enough detail to be checked, and marks each claim with its epistemic status:

Status	Meaning
kernel	proved in Lean 4; in `AxiomAudit.lean`; axioms within budget
kernel (witness)	a structure is exhibited and its fields are proved in Lean
script	verified by exhaustive computation in Python (`docs/*.py`), not formalised
conjecture	stated in `PLAN.md`, no proof

4.1 The formal signature

The library separates the ring from the category. `FusionRing n` carries $N\colon \operatorname{Fin} n \to \operatorname{Fin} n \to \operatorname{Fin} n \to \mathbb{N}$, the involution `dual`, the unit at 0, associativity, and the two duality axioms. `FusionSystem n k` extends it with the categorical data:

Field(s)	Content
`mult_free`	$N_{ij}^k \le 1$
`F`, `F_support`	the associator $F_d^{abc}(e,f)$, vanishing off the channels
`pentagon`	the ten-label pentagon
`F_unit_left`, `F_unit_right`	unit normalisations at $a=1$ and $c=1$
`Finv`, `Finv_support`, `Finv_comp`, `F_comp_inv`	the inverse associator and its two inverse laws
`ev`, `coev`, `ev_ne_zero`, `coev_ne_zero`	the duality scalars
`snake_left`, `snake_right`	the two snake equations
`J`, `J_ne_zero`, `J_unit_left`, `J_unit_right`, `J_dual`, `J_coherent`	the double-dual twist and its four laws

A pivotal structure is then a family $p\colon \operatorname{Fin} n \to \Bbbk^\times$ with $p_0 = 1$ and $p_k = J_{ij}^k p_i p_j$ on every fusion channel; the dimension is $\dim_i = p_i \kappa_i$ with $\kappa_i = \mathrm{ev}_i \mathrm{coev}_i$; sphericity is $\dim_{i^*} = \dim_i$.

The one design decision that deserves comment is `Finv`. The associator at fixed (a,b,c,d) is a square matrix whose rows are the channels of $b \otimes c$ and whose columns are the channels of $a \otimes b$; these are *different* subsets of $\operatorname{Fin} n$, so there is no canonical way to present the inverse as a matrix indexed by $\operatorname{Fin} n$. The project therefore carries α^{-1} as data with its two inverse laws. This is not pedantry: the right snake equation needs the entry $(\alpha^{-1})^{(0,0)}$, and getting this wrong excluded Fibonacci and Ising from the formalism (§4.3).

4.2 Four defects, and what caught them

	Symptom	Diagnosis	Regression test
D1	every theorem vacuous	unit axioms inconsistent at $(0,b,0,d,b,0)$ for $n \ge 2$	`VecZ2.system`
D2	"pentagon" fails on $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$, all ω	eight free labels instead of ten	512-tuple check
D3	target provable with no hypothesis	`p_fusion` fixed $J \equiv 1$	`VecZ2Twisted` witness with $\kappa_g = -1$
D4	Fibonacci, Ising uninhabitable	reciprocal used for matrix inverse in `snake_right`	Fib, Ising

D1–D3 share a signature: a statement that *looks* like its source but quantifies over an empty or over-constrained type. D4 is different — a genuine mathematical error in a normalisation — and

it is the one that took three attempts (§4.3). The lesson recorded in the plan is that "the build is green" certifies nothing about the statement; only inhabitation and an independent computation do.

4.3 The duality normalisation

Theorem 4.1 (Snake normalisation; kernel)**.** *Let $\kappa_i = \mathrm{ev}_i \,\mathrm{coev}_i$. In a skeletal rigid datum the two snake equations are equivalent to*

$$\kappa_i \, F_i^{i\,i^*\,i}(0,0) = 1 \qquad \text{and} \qquad \kappa_i \, \left(F_{i^*}^{i^*\,i\,i^*}\right)^{-1}(0,0) = 1,$$

where the second factor is the $(0,0)$-entry of the matrix inverse *of the associator block, not the reciprocal of the single entry $F_{i^*}^{i^*ii^*}(0,0)$.*

The two readings agree exactly when the block is 1×1 — true for every pointed category, which is why the earlier "κ/F" correction sufficed for $\mathrm{Vec}^{\omega}_{\mathbb{Z}/3}$ — and differ as soon as a self-dual simple has a 2×2 block. Fibonacci is the minimal case: with the gauge $F_\tau^{\tau\tau\tau} = \left(\begin{smallmatrix} 1/\varphi & 1 \\ 1/\varphi & -1/\varphi \end{smallmatrix}\right)$ the left snake gives $\kappa_\tau = \varphi$, the reciprocal reading demands $\kappa_\tau = 1/\varphi$, and the matrix-inverse reading gives $\varphi \cdot (1/\varphi) = 1$ because the block is an involution. The library proves that the wrong reading fails (`Fibonacci.swapped_Finv_comp_ne_one`), so the correction is itself regression-tested.

Consequences, all kernel-checked:

- `FibonacciSystem : FusionSystem 2` and `IsingSystem : FusionSystem 3`, both over $\mathbb{R}$, with every field proved: pentagon, both unit laws, both inverse laws, both snakes;
- the pivotal structure $p \equiv 1$ is spherical for both, with $\dim = (1, \varphi)$ and $\dim = (1, \sqrt{2}, 1) = \mathrm{FPdim}$;
- $\mathrm{Vec}_{\mathbb{Z}/2}$ still inhabits the structure, so the correction did not buy the examples at the cost of the trivial case;
- the finite checks are done by *exact* arithmetic in $\mathbb{Z}$: each entry is an integer pair (p, q) denoting $(p + q\sqrt{d})/2$ in $\mathbb{Q}(\sqrt{d})$, and `decide` verifies the identities. There is no `norm_num` over the reals and no `native_decide`.

The cost is compile time: the file takes about 100 seconds, so the package build is about three minutes.

Remark 4.2 (Statement of the snake in the general case). The same computation in a non-skeletal category produces two extra structural maps (the duality comparison and the unit constraint). The skeletal form above is the image of the general snake under the equivalence, and the correspondence is the content of WP5's bridge; it is *not* obvious and should not be assumed silently.

4.4 The double-dual twist, and the two identities

Theorem 4.3 (Pivotal dimension is a coboundary; kernel)**.** *Let P be a pivotal structure and define the normalised twist*

$$\tau_{ab}^{c} \;=\; J_{ab}^{c} \cdot \frac{\kappa_c}{\kappa_a \kappa_b}.$$

Then for every fusion channel,

$$\dim_P(c) \;=\; \tau_{ab}^{c} \, \dim_P(a) \, \dim_P(b).$$

Consequently $\dim_P$ *is multiplicative on the fusion ring if and only if*

$$\sum_c N_{ab}^{c} \, J_{ab}^{c} \, \kappa_c \;=\; \kappa_a \kappa_b \qquad \textit{for all } a, b \qquad \textbf{(M)} \tag{6}$$

and ENO 8.23 is equivalent to the per-channel identity

$$J_{ab}^{c} \;=\; \frac{f_c \, \kappa_a \kappa_b}{\kappa_c \, f_a f_b} \qquad \textit{on every channel with } N_{ab}^{c} > 0. \qquad \textbf{(K)} \tag{7}$$

The equivalence is two theorems in the library, not one: `eno823_of_twistNorm` proves (K) ⇒ ENO 8.23, and `twistNorm_eq_fpdim_ratio_of_dim_eq` proves the converse. Together with `mult_iff_twist` ((M) ⇔ multiplicativity of $\dim_P$) they make the target a statement about (N, J, κ) alone.

Two consequences deserve emphasis.

(i) **(K) is not invariant under the vertex gauge.** Rescaling $\mathrm{coev}_i \mapsto \lambda_i \mathrm{coev}_i$, $\mathrm{ev}_i \mapsto \mathrm{ev}_i/\lambda_i$ preserves the snakes and fixes κ, but multiplies the defect $\Delta^c_{ab} = J^c_{ab}\kappa_c f_a f_b/(f_c \kappa_a \kappa_b)$ by $t_a t_b / t_c$ with $t_i = \lambda_i/\lambda_{i^*}$. So (K), hence the target, survives only the duality normalisation $\lambda_{i^*} = \lambda_i$. The repair is to add $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ as an axiom, or to define $\dim_i = p_i \,\mathrm{coev}_i\, \mathrm{ev}_{i^*}$. This is a defect of the signature, not of the mathematics, and WP1 fixes it.

(ii) **On the coevaluation channel (K) is J-free.** On $(i, i^*; 1)$ the axiom $J^1_{ii^*} = 1$ applies, so (K) reads

$$\kappa_i \, \kappa_{i^*} = f_i^2 \quad \text{for all } i, \qquad \textbf{(C)} \tag{8}$$

which is exactly sphericity of $p = f/\kappa$. Hence (K) ⇒ (C) is a two-line theorem, while (C) ⇒ (K) is ENO 8.23. The residual mathematical content is therefore *pseudo-unitarity* ⇒ (C): a statement about κ and f only, with no J in it.

4.5 Refutations

Theorem 4.4 (Yang–Lee; kernel)**.** *There is a complete skeletal rigid datum* `YangLee.system` *over* $\mathbb{R}$ *— Fibonacci's fusion ring, associator* $F^{\tau\tau\tau}_\tau = \left(\begin{smallmatrix} -\varphi & 1 \\ -\varphi & \varphi \end{smallmatrix}\right)$, $\mathrm{ev} \equiv 1$, $\mathrm{coev}_\tau = -1/\varphi$, *whence* $\kappa = (1, -1/\varphi)$ *— satisfying the pentagon on all* 2^9 *label tuples, both unit laws, both inverse laws and both snakes, such that*

$$\neg\, \textit{ENO823 YangLee.system FibonacciFPData}.$$

Indeed $\kappa_\tau \kappa_{\tau^*} = 1/\varphi^2 \neq \varphi^2 = f_\tau^2$, *and on the channel* $(\tau, \tau; 1)$ *the axiom* $J^1_{\tau\tau} = 1$ *forces* $1 = \varphi^4$ *for any pivotal structure with* $\dim = \mathrm{FPdim}$.

Three things follow, and they shape the whole programme.

(a) ENO 8.23 is **false** as an implication from the axioms of a rigid fusion datum. The missing hypothesis is pseudo-unitarity, and no condition on J can replace it: the contradiction uses only $J^1_{\tau\tau} = 1$, which is an axiom.

(b) The counterexample is *two-simple*. There is no hope that a rank threshold excludes it, and any formalisation that omits positivity will silently prove a false theorem — which is what the first version of this project did.

(c) Yang–Lee is not pseudo-unitary, which is exactly ENO's own remark. The formal statement `YangLee.not_pseudoUnitary` records the numerical failure $\sum_i \kappa_i \kappa_{i^*} = 1 + \varphi^{-2} \neq 1 + \varphi^2$.

Theorem 4.5 (Pivotal structures are not ring-theoretic; kernel (witness))**.** $\mathrm{Vec}^\omega_{\mathbb{Z}/2}$ *with the non-trivial cocycle (*$F = -1$ *on the* $g \otimes g \otimes g \to g$ *block, duality normalisation* $\kappa_g = -1$*) is a* `FusionSystem 2` *over* $\mathbb{R}$*, carrying two pivotal structures:* $p \equiv 1$*, with* $\dim g = -1$*, and* $p_g = -1$*, with* $\dim g = 1$*, which is spherical. The conclusion of ENO 8.23 holds for the second and fails for the first.*

Theorem 4.5 is the counterexample to "the dimension is a function of the fusion ring". It also shows why pseudo-unitarity is the right hypothesis rather than a technical convenience: it selects, among the pivotal structures, the one with positive dimensions.

Remark 4.6 (A third witness, at script level). For $\mathrm{Vec}^\omega_{(\mathbb{Z}/2)^2}$ with $\omega = (-1)^{a_1 b_2 c_1}$ one has $\kappa = (1, 1, 1, -1)$, which is not a character of $(\mathbb{Z}/2)^2$. With $J \equiv 1$, (M) fails; with $J = \delta(\kappa) = \kappa_a \kappa_b / \kappa_c$ it holds. This computation is in `docs/pointed_twist_check.py` and is *not* yet kernel-checked. It is the reason the signature must not add $J^2 = 1$ as an axiom: for $\mathrm{Vec}^\omega_{(\mathbb{Z}/3)^2}$ with $\omega = \zeta^{a_1 b_2 c_1}$ the honest twist squares to a primitive cube root of unity, and that category is unitary, so such an axiom would exclude a category to which the theorem applies.

4.6 ENO Proposition 8.21

Theorem 4.7 (ENO 8.21; kernel, modulo `DimEigen`)**.** *Let R be a fusion ring with `FPData` and let $d\colon I \to \mathbb{R}$ satisfy $\sum_j N_{ij}^k d_j = d_i d_k$ for all i, k. Then $|d_i| \le f_i$ for all i. If moreover $\sum_i |d_i|^2 = \sum_i f_i^2$ then $|d_i| = f_i$ for all i.*

The proof formalised is ENO's: the weighted norm $\|x\| = \sum_j f_j |x_j|$ is submultiplicative on the eigenvector equation, and `fpdim_char` — the fact that f is a *character*, $f_i f_j = \sum_k N_{ij}^k f_k$ — bounds it by f_i. Two index conventions occur (N_{ij}^k against N_i^{jk}), and the library proves the cyclic symmetry $N_{ij}^k = N_{jk^*}^i$ needed to pass between them (`cyclic`, `cyclic1`, `cyclic2`); the repo's axioms do not otherwise connect the second and third slots.

The equality case is `eq_abs_of_sum_sq`, and `abs_dim_le_fpdim_of_pseudoUnitary` discharges `DimEigen` from multiplicativity, so ENO's $|V|^2 = \mathrm{FPdim}(V)^2$ is available for any pivotal structure whose dimension is multiplicative.

Remark 4.8 (What 8.21 does not give). It is tempting to think that the T-matrix route bypasses multiplicativity. It does not. ENO's Remark 3.1 says that in the pivotal case the signs ε_i and b_j^{ik} are 1, so $T_j^{ik} = N_j^{ik}$ and the eigenvector equation $T_i d = d_i\, d$ *is* multiplicativity of the dimension. The library proves this identification (`tMatrix_mulVec`, `abs_tMatrix`: the domination $|(T_i)_{kj}| \le N_j^{ik}$ is an equality). The non-trivial b-symbols exist only when no pivotal structure is fixed, and their role is to prove $\dim \mathcal{C} \neq 0$ without one.

4.7 ENO's own route: the square root a and the b-cochain

ENO's proof does not start from a pivotal structure — it produces one — so a signature that begins with p cannot express it. The library therefore also carries ENO's input.

Definition 4.9 (kernel, `SqrtData`)**.** *A* square root of g *is an additive isomorphism $a\colon \mathrm{Id} \to (-)^{**}$ with $a^2 = g$ monoidal, where $g\colon \mathrm{Id} \to (-)^{****}$ is ENO Theorem 2.6. In a skeleton, monoidality of a^2 has to be stated against J^2, because on labels $(-)^{**}$ is the identity: the tensorator of $(-)^{****} = (-)^{**} \circ (-)^{**}$ is J^2, not J.*

Given such an a, ENO set

$$b_c^{ij} \;=\; J_c^{ij}\,\frac{a_i a_j}{a_c}, \qquad T_c^{ij} \;=\; N_c^{ij}\, b_c^{ij}, \qquad d_i \;=\; \mathrm{Tr}(a_i) \;=\; a_i \,\mathrm{coev}_i\, \mathrm{ev}_{i^*}.$$

Theorem 4.10 (kernel)**.** *With these definitions:*

(i) $(b_c^{ij})^2 = 1$ *on every channel, and it follows from $a^2 = g$ monoidal, not from $J^2 = 1$ (`b_sq`);*

(ii) the trace identity $d_i d_j = \sum_c T_c^{ij} d_c$ is equivalent *to*

$$\kappa'_i \kappa'_j = \sum_c N_c^{ij}\, J_c^{ij}\, \kappa'_c, \qquad \kappa'_i = \mathrm{coev}_i\, \mathrm{ev}_{i^*},$$

which is (M) with the duality normalisation κ' (`trace_iff`);

(iii) $d_i = a_i \kappa'_i$, and $d_i = a_i \kappa_i$ only when $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ (`d_eq_kappa`).

The content of Theorem 4.10 is an identification, not a discovery: ENO's "by the definition of b" step is (M). But the identification is what dissolves an apparent fork in this project's plan. "Derive J" and "assume multiplicativity" are not two research programmes; they are two faces of one theorem, and the condition J has to satisfy is (M) itself.

Remark 4.11 (What must not be assumed). `Cover.lean`'s `S4Data.g_fusion` reads $g_c = g_i g_j$ on channels. That is the $J = 1$ special case and is wrong in general; it must be restated against J^2. Likewise $J^2 = 1$ and $b^2 = 1$ must not be added as axioms (§4.5).

4.8 The instances, and the audit

Datum	Ring	$\mathbb{k}$	κ	Role
`VecZ2.system`	$\mathbb{Z}/2$	$\mathbb{R}$	$(1,1)$	inhabitant; pentagon on 512 tuples
`VecZ2Twisted.system`	$\mathbb{Z}/2$, $\omega \neq 1$	$\mathbb{R}$	$(1,-1)$	two pivotal structures
`FibonacciSystem`	Fib	$\mathbb{R}$	$(1,\varphi)$	dim = FPdim, spherical
`IsingSystem`	Ising	$\mathbb{R}$	$(1,\sqrt{2},1)$	dim = FPdim, spherical
`k2210`	$K(1,0,1)$	$\mathbb{R}$	—	`FPData` certificate; FPdim $= (1,2,1)$, $D^2 = 6$
`YangLee.system`	Fib	$\mathbb{R}$	$(1,-1/\varphi)$	refutes ENO823 without hypotheses

`AxiomAudit.lean` runs `#print axioms` on 73 declarations covering every result quoted above and reports `[propext, Classical.choice, Quot.sound]` throughout. The package builds in 3,032 jobs with warnings only. There is no `sorry` in the library (the single occurrence of the string is in a comment recording the axiom budget).

`ENO823` itself is a `def`, not a `theorem`: it records the open target. The conditional theorem `eno823_of_pseudoUnitary` takes positivity and multiplicativity of a chosen pivotal dimension as hypotheses (`IsPseudoUnitaryPivotal`) together with `PFUnique`, and concludes the theorem. Both hypotheses are necessary: positivity fails on Yang–Lee, and multiplicativity fails on $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$.

4.9 The exact gap

Stated as a list, the gap between the present library and the target is:

(A) **The twist is data, not derived.** J is constrained by four laws and no closed form is known for general multiplicity-free data. Consequences: `p_mon` mentions an undetermined object, and (M) does not follow from the four twist laws as such — a datum may carry an arbitrary coherent J, and on $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$ it fails — so (M) has to come from the derivation of J, not from the axioms.

(B) **The analytic step is absent.** Positivity of a dimension function on a pseudo-unitary category, and the two twists $\sigma = d^2/|d|^2$, $\tau = d/|d|$ that ENO use to normalise it, are not formalised.

(C) **The signature is narrower than the mathematics.** No triangle axiom; no labels with $i^{**} \neq i$; multiplicity-freeness assumed; $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ not assumed (so (K) is gauge-fragile).

(D) **The cover and grading layer is scaffolding.** `coverGlobalDim_double` is a reindexing identity on $\sum d^2$; `coverCanonicalPivotal` returns the input scalar with no `PivotalStructure`; the three former axioms of `Grading.lean` are gone and their content is a theorem plus two `def`s recording open targets.

(E) **No bridge to Mathlib.** `MathlibPatches/` is a stub, and `FPData` is supplied by hand.

These five items are the work packages of §5.

5 Research programme

5.1 Three aims, and the logical geography of the target

The programme has three aims, corresponding to Q1–Q3, and one exploratory strand for Q4.

A1 (Signature).
A formal signature for fusion data that is inhabited, faithful, and provably equivalent to the abstract notion: triangle axiom, general double-dual permutation, multiplicities, gauge-invariant duality normalisation, and a bridge to Mathlib's `PivotalCategory`.

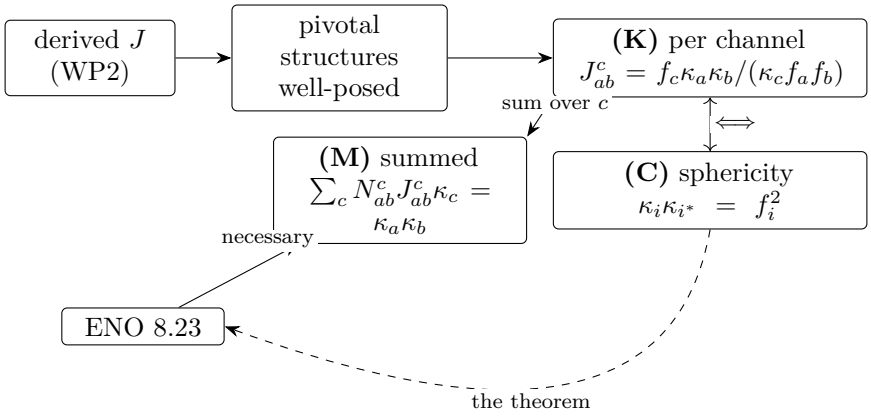


Figure 1: The logical geography. (K), (C) and ENO 8.23 are mutually equivalent: (K)⇒(C) is a two-line computation on the coevaluation channel, (C)⇒(K) is the theorem, and (K) is the coboundary form of the target. (M) is strictly weaker — Yang–Lee satisfies it and fails the target — so ENO 8.23 ⇒ (M) is the only implication available between them. The derivation of J is logically prior: it makes the other boxes statements about determined objects. Dashed: the direction that carries the analytic content.

A2 (Twist).
The double-dual twist J derived from $(F, \mathrm{ev}, \mathrm{coev})$, with its four laws and its general pentagon proved, so that "p is pivotal" becomes a well-posed predicate rather than a constraint on undetermined data.

A3 (Theorem).
A formal proof of ENO 8.23 in the exact form of Theorem 2.8: existence, sphericity, positivity, equality with FPdim, and uniqueness.

A4 (Frontier, exploratory).
The pivotal cover, the universal grading group, and the 2-torsion obstruction: what can be said toward ENO Conjecture 2.8.

The logical relations between the statements are worth recording once, because the project's own plan confused them for several weeks. Figure 1 is the map.

Three readings of the figure matter for planning. First, **(M) is not the target**: it is multiplicativity of whichever dimension function is at hand, it holds for Yang–Lee, and it is strictly weaker than (K). Second, **(K) and (C) are the target**, and the content of the theorem is the passage from pseudo-unitarity to (C) — an equation in κ and f alone, with no J. Third, **the derivation of J does not prove anything about the target**; it removes an undefined object from the statement. This is why A2 and A3 are separate work packages and why neither can be skipped.

5.2 Work packages

WP0 — Foundations and reproducibility (months 0–4)

Objective. Replace hand-supplied inputs with proved ones, and make the build reproducible from a clean checkout.

Tasks.

0.1 Vendor the Perron–Frobenius chain (#39914–#39922) and define FPdim as the Perron root of the fusion matrix; prove Theorem 2.3 (existence, positivity, uniqueness, character form

(1), integrality) and delete the `FPData` hypothesis from every theorem that can do without it. Where a theorem genuinely needs uniqueness, replace `FPData` by the weaker `PFUnique` already isolated in `PFDim.lean`.

0.2 Clean-tree CI: a fresh clone, `lake exe cache get`, `lake build`, and `lake env lean AxiomAudit.lean` as three separate gates. At present a from-clean build of the committed tree has never been verified — earlier builds used a path override — and that is a reproducibility debt no deliverable should inherit.

0.3 A per-file build harness with a timeout, so that a 100-second example file does not block work on the core.

0.4 `BLUEPRINT.md` with the module dependency graph (Figure 2) and a statement-to-source table: every theorem in the library mapped to the ENO/EGNO/Bartlett statement it formalises, with the numbering convention recorded.

Deliverable. D1, D2. **Success criterion.** A machine with the toolchain and no network builds the library and passes the audit from a clean clone.

WP1 — Hardening the signature (months 1–10)

Objective. A1: make the finite-data signature express the mathematics, not a special case of it.

Tasks.

1.1 **Triangle axiom.** Add $F_d^{a\,1\,c}(a,c) = \delta_{d,\dots}$, the one axiom of Mac Lane's list missing from the signature, and discharge it for every instance. It is independent of the two unit laws already present.

1.2 **The double-dual permutation.** Introduce the label permutation $\sigma\colon \mathrm{Fin}\, n \to \mathrm{Fin}\, n$ with $\sigma i = i^{**}$ and $\sigma^2 = \mathrm{id}$, restate the twist coherence in its general form

$$J_{ab}^{e}\, J_{ec}^{d}\, F_{d}^{abc}(e,f) \;=\; J_{bc}^{f}\, J_{af}^{d}\, F_{\sigma d}^{\sigma a\, \sigma b\, \sigma c}(\sigma e, \sigma f),$$

and prove that the F-free form currently used is its $\sigma = \mathrm{id}$ degeneration. A word of caution that must go into the blueprint: *a model with $\sigma \neq \mathrm{id}$ is a non-pivotal fusion category*, hence a counterexample to ENO Conjecture 2.8. The general branch therefore cannot be tested by an example; it can only be tested by the degeneration and by the reconstruction theorem of WP5. This is a rare case in formalisation where the general statement has no inhabitant that we would be willing to exhibit.

1.3 **Gauge-invariant duality normalisation.** Either add $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ as an axiom, or redefine $\dim_i = p_i\, \mathrm{coev}_i\, \mathrm{ev}_{i^*}$ as ENO do. The two choices have different consequences: the first makes (K) gauge-invariant and kills a family of false negatives; the second matches the source but changes every instance proof. The proposal is to do the first, and to prove the equivalence of the two normalisations in the self-dual case.

1.4 **Multiplicities.** Replace the scalar $F_d^{abc}(e,f)$ by a family of matrices indexed by channel bases, $\left(F_d^{abc}\right)_{(e,\alpha),(f,\beta)}$, with the pentagon stated on matrix products. This is a separate development (§6.4); it must not be attempted by patching the multiplicity-free file, because every instance proof depends on `decide` over `Fin n` and would silently keep the scalar reading.

Deliverables. D3, D9. **Success criterion.** All six existing instances re-verified under the hardened signature, with the same axiom budget, and the three new axioms proved rather than assumed.

WP2 — Deriving the double-dual twist (months 2–14)

Objective. A2: J as a function of $(F, \mathrm{ev}, \mathrm{coev})$.

Tasks.

2.1 **The comparison map.** For each pair (a, b) define the cuspidal comparison $\psi_{a,b}\colon (a \otimes b)^* \to b^* \otimes a^*$ as the unique map intertwining the evaluations, and prove it is an isomorphism. This is a finite computation in the skeleton: two F-moves and one transported coevaluation.

2.2 **The ratio formula.** Prove

$$J_{ab}^{c} \;=\; \frac{\psi_{a,b}|_{c}}{\psi_{b^*,a^*}|_{c^*}},$$

and, in the normalisation $\mathrm{ev}_{i^*} = \mathrm{ev}_i$, the closed form

$$\Omega(a, b; c) = F_0^{b^* a^* c}(c^*, b) \cdot (M_b)^{-1}(0, c), \qquad M_b(E, c') = F_b^{a^* a b}(E, c'),$$

and then

$$J_{ab}^{c} = \Omega(b^*, a^*; c^*)/\Omega(a, b; c),$$

where M_b is restricted to its support (unrestricted it is rectangular — for Ising at $(a, b) = (\sigma, \psi)$ it is 2×1 — and the formula returns $0/0$), and $(M_b)^{-1}$ is a genuine matrix inverse.

2.3 **The four laws.** Prove $J_{ab}^{0} = J_{i^*0}^{i} = 1$, non-vanishing, and the general pentagon of Task 1.2. This is the step that turns "J is data" into "J is computed", and it is independent of pseudo-unitarity: it holds for Yang–Lee too.

2.4 **Independent confirmation.** Compare with Bartlett's Theorem 4.17 (§3) index by index, and with the sixteen-data-set census already in `docs/`. A mismatch in either is a defect in the derivation, not in the check.

Deliverables. D4. **Success criterion.** The library contains no hypothesis on J beyond the four laws, and those four laws are theorems about the derived object. **Explicit non-goal:** WP2 does not prove (K). Deriving J turns (M) into a statement about a determined object; since the dimension of a pivotal category is multiplicative, (M) should then be a theorem, and proving it is a further acceptance test of the derivation. But the target is the strictly stronger per-channel identity (K), which fails for Yang–Lee, and that is WP3's content.

WP3 — The analytic core: from pseudo-unitarity to positivity (months 4–20)

Objective. A3: ENO 8.23.

Tasks.

3.1 **Construct a.** ENO choose $a_V\colon V \to V^{**}$ with $a_V^2 = g_V$, and the content is the coherence: the monoidality defect b of a must satisfy $(b_c^{ij})^2 = 1$. Prove existence of such an a from the monoidal isomorphism g (Theorem 2.10), which is the hypothesis the project takes. `SqrtData` currently *assumes* a; this task constructs it.

3.2 **The bound.** From the trace identity $d_i d_j = \sum_c T_c^{ij} d_c$ with $T_c^{ij} = N_c^{ij} b_c^{ij}$ and $|b| = 1$, derive $|d_i| \le f_i$ by ENO's weighted norm. The library has the weighted-norm argument (`abs_le_fpdim_of_eigenvector`) and the domination; what is missing is the passage from the T-matrix to the N-matrix, which is the triangle inequality $|\sum_j T_j^{ik} d_j| \le \sum_j N_j^{ik} |d_j|$.

3.3 **The sum identity.** Prove $\sum_i |d_i|^2 = \dim \mathcal{C}$ for $d_i = a_i \kappa_i'$, which under pseudo-unitarity gives $\sum_i |d_i|^2 = \sum_i f_i^2$; the equality case of Theorem 4.7 then yields $|d_i| = f_i$.

3.4 **The two twists.** Show that $\sigma_i = d_i^2/|d_i|^2$ and $\tau_i = d_i/|d_i|$ are tensor autoequivalences of the skeletal datum (they act by signs on simples and preserve F, ev, coev), and that twisting a by them produces a' with $d' = f > 0$. Then `toPivotal` produces the pivotal structure with dim = FPdim.

3.5 **Uniqueness and positivity.** Assemble `eno823_exact`: existence, positivity, equality with FPdim, and (∃!) uniqueness, using `pivotal_ext` and `PFUnique`. The conditional form `eno823_of_pseudoUnitary` already exists and is the checkpoint this task must reach unconditionally.

Deliverables. D5, D6. **Success criterion.** `ENO823Exact` proved with no hypothesis other than `PFUnique` (or none, once WP0 discharges it), and `YangLee.not_pseudoUnitary` as the sharpness witness.

WP4 — Cover, grading, and the frontier (months 8–18)

Objective. A4: the part of the theory that does not need pseudo-unitarity, stated precisely enough to say what is missing for ENO Conjecture 2.8.

Tasks.

4.1 **The cover.** For $g\colon \mathrm{Id} \to (-)^{****}$ monoidal, define

$$\widetilde{\mathcal{C}} = \{(X, f) \mid f\colon X \to X^{**},\ f^{**} f = g_X\}$$

with the induced tensor structure; prove it is a fusion category, $\mathbb{Z}/2$-graded over $\mathcal{C}$, that the forgetful functor preserves $|\cdot|^2$, that it carries a canonical pivotal structure $p_{(X,f)} = f$, and that $\dim \widetilde{\mathcal{C}} = 2\dim \mathcal{C}$, $\mathrm{FPdim}\widetilde{\mathcal{C}} = 2\mathrm{FPdim}\mathcal{C}$. This is ENO Remark 3.1 and Bartlett §5.3; the present file (`Cover.lean`) proves only a reindexing identity on $\sum d^2$ and returns the scalar f without a `PivotalStructure` — it is scaffolding, and the blueprint must say so.

4.2 **The torsor.** Prove that pivotal structures form a torsor under $\mathrm{Hom}(U_{\mathcal{C}}, \Bbbk^\times)$, where $U_{\mathcal{C}}$ is the universal grading group, and identify the class of g in it.

4.3 **The splitting criterion.** Prove

$$\mathrm{Nonempty}\,(\mathrm{PivotalStructure}\,\mathcal{C}) \iff \exists\, h,\ h^2 = [g],$$

and deduce that $\mathcal{C}$ is pivotal whenever $U_{\mathcal{C}}$ has no 2-torsion. This is the known partial result toward Conjecture 2.8 and it is entirely algebraic.

4.4 **What the formalisation adds here.** The obstruction class $[g]$ is usually handled informally; a formal statement of the criterion would make precise which fusion rings are excluded by 2-torsion and which Frobenius–Schur indicator patterns are compatible with sphericity. No new mathematics is promised in this package — the value is precision.

Deliverables. D7. **Success criterion.** The cover is a `FusionSystem` with a `PivotalStructure`, and the torsor statement is proved, not assumed.

WP5 — The categorical bridge and upstream (months 10–24)

Objective. A1, second half: a skeletal datum *is* a fusion category, and the pivotal structure is Mathlib's.

Tasks.

5.1 **The category of a datum.** Objects: formal finite direct sums of simples. Morphisms: matrices of scalars indexed by channels. Prove it is $\Bbbk$-linear, abelian, semisimple, monoidal, rigid, with $\mathrm{End}(\mathbf{1}) = \Bbbk$ — the reconstruction theorem. This is the largest single piece of engineering in the programme and the one most exposed to Mathlib churn.

5.2 **The pivotal equivalence.** Prove PivotalStructure $S \simeq$ PivotalCategory (CategoryOf S) as a bijection of data, i.e. that Mathlib's monoidal natural isomorphism $\mathrm{Id} \to (-)^{**}$ restricts to the scalars p_i and that `p_mon` is its coherence.

5.3 **Delete the patch.** When #42145/#42150/#42191 merge, replace the vendored `PivotalCategory.lean` by an import; keep the file name as a re-export for one release and then remove it. If #42191 stalls, contribute the spherical layer upstream ourselves, since the definition is short and the present library has the examples to test it.

5.4 **Contribute back.** The Perron–Frobenius material and the spherical layer are the two pieces of independent value; both should go upstream with the examples as tests.

Deliverables. D8, D10. **Success criterion.** `MathlibPatches/` contains no mathematics, only imports.

WP6 — Instances, witnesses, and the regression suite (continuous)

Objective. Every theorem in the library has an inhabitant, and every defect in Table 1 has a regression test.

Tasks.

6.1 Extend the instance set: $\mathrm{Vec}^{\omega}_{\mathbb{Z}/n}$ for $n = 2, \dots, 6$; Tambara–Yamagami for $\mathbb{Z}/2$; the rank-three ring $K(1, 0, 1)$ already present; H_3 (non-integer FPdim) as a test that nothing assumes integrality.

6.2 Formalise the two script-level witnesses of §4.5: $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$ (the failure of (M)) and $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/3)^2}$ (the failure of $J^2 = 1$). The first is the kernel-checked answer to "why can't (M) be an axiom?".

6.3 A multi-multiplicity datum: the rank-two based ring $X^2 = \mathbf{1} + 2X$, purely algebraically, to exercise the matrix-valued associator of WP1 Task 1.4.

6.4 A census harness: run the derived J against the Python reference computations for every data set, and fail the build on disagreement. Disagreement between Lean and the Python scripts has already caught two transcription errors, one in each direction.

Deliverables. D9, D11. **Success criterion.** Sixteen data sets, at most two of them script-only, and the census harness green.

WP7 — Dissemination (continuous)

Three papers, a blueprint, and an artifact: see §9. The dissemination order is deliberate — the negative results first, because they are short, self-contained, and they protect the reader of the main paper from the statement without hypotheses.

5.3 Milestones

Milestone	Month	Content	Deliverable
M1	4	`FPData` discharged; clean-tree CI; blueprint	D1, D2
M2	8	Signature v2 (triangle, σ-twist, gauge fix); six instances re-verified	D3
M3	12	J derived; four laws and general pentagon proved; Bartlett comparison	D4
M4	16	T-matrix domination; bound $\lvert d_i\rvert \le f_i$ from a; sum identity	D5
M5	20	Positivity and σ/τ twists; `ENO823Exact` proved	D6
M6	22	Cover and torsor; bridge to `PivotalCategory`; patch deleted	D7, D8
M7	24	Release v1.0; three papers submitted; artifact archived	D10–D12

6 Techniques

6.1 Finite data and exact arithmetic

Everything categorical is reduced to finite algebra over a field, and everything numeric is done exactly. Two devices carry the instances.

Integer pairs in $\mathbb{Z}[\sqrt{d}]$. Real numbers never appear in a proof obligation. Each entry of F, ev, coev, J in the Fibonacci, Ising and Yang–Lee examples is an integer pair (p, q) denoting $(p + q\sqrt{d})/2$ in $\mathbb{Q}(\sqrt{d})$; the field operations are defined on pairs, and `decide` checks the pentagon, the unit laws, the inverse laws and the snakes on all label tuples. For Ising, $d = 2$ and the scalar $1/\sqrt{2}$ enters as a normalisation, so the identities are checked twice — once for the rational part and once for the $\sqrt{2}$ part. This is why the examples are trustworthy despite being numerical, and it is why they compile in 100 seconds rather than hours.

Field checks are hand-written. Closing real-number arithmetic is not automatic: `one_lt_pow` and its primed variant were unusable in the Yang–Lee file, and the φ^4 contradiction had to be assembled by hand from `nlinarith` with an explicit $\sqrt{5} \cdot \sqrt{5} = 5$. A mis-stated closed form for $((1 + \sqrt{5})/2)^2$ cost a build cycle. The practical rule adopted is: get the closed form right on paper, then prove it once, then reuse.

6.2 Inhabitant-first statement design

The four defects of §4 share one cause: a statement was written to look like its source and then trusted. The remedy, now institutionalised in this project, is that *no statement is committed until an inhabitant exists*, and no axiom is believed until an example satisfies it and a non-example does not. Concretely:

(i) every structure gets a witness in the same commit as its definition (`VecZ2.system` with the definition of `FusionSystem`);

(ii) every axiom gets a regression test that fails if the axiom is strengthened (`swapped_Finv_comp_ne_one` for the snake; `VecZ2Twisted` for the twist);

(iii) every theorem whose statement mentions a hypothesis gets a witness that the hypothesis is needed (Yang–Lee for positivity; $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ for multiplicativity).

The cost is real — the witness files are about half the library — and it is the single highest-value investment in the programme, because the defects it catches are exactly the ones that survive refereeing.

6.3 Falsification before proof

For each proposed statement S, the programme first looks for a counterexample, in this order: (i) on paper, on the smallest rank; (ii) in Python, on the full data set, with the associators composed as matrices rather than manipulated symbolically; (iii) in Lean, by attempting the statement on the counterexample candidate before attempting the theorem.

This order is not a heuristic. Two of the project's four defects were found this way, and a third — the '$J \equiv 1$' claim for pointed categories — was refuted by an exhaustive computation in `docs/pointed_twist_check.py` after it had been stated in the plan. The Python step matters because matrix composition over channel bases is where index conventions break, and because a script that recomputes every scalar from the snake axioms is an independent implementation, not a restatement.

6.4 Multiplicities: why this is a separate development

With $N_{ij}^k > 1$ the Hom space $\mathrm{Hom}(X_k, X_i \otimes X_j)$ has dimension N_{ij}^k and the associator at fixed (a, b, c, d) is a matrix whose entries are themselves matrices; the pentagon becomes a matrix identity with channel bases on both sides, and "the" F-symbol is a choice of bases. Three consequences:

(i) the scalar signature cannot be extended by adding a parameter — the type of `F` changes, and every instance proof that ends in `decide` over `Fin n` must be rewritten;

(ii) gauge freedom is larger (a GL_N-valued change of channel bases), so the twist J becomes a matrix between channel spaces and the ratio formula of WP2 needs a determinant or a trace to become scalar;

(iii) the correct target in the general case is the *categorical* statement (WP5), with the skeletal statement as a corollary in the multiplicity-free case.

The proposal therefore treats the multiplicity-free theorem as the deliverable and the general case as a stretch goal, pursued through the categorical bridge rather than through a bigger scalar signature. This is a scope decision, stated here so that it can be argued with.

6.5 Diagram evaluation and independent reference computations

The one place where the project has repeatedly gone wrong is the evaluation of a diagram with more than two F-moves. The technique that works is to evaluate the same diagram twice by different routes:

- an *evaluation-side* assembly (compose evaluations, transport coevaluations);
- a *coevaluation-side* assembly;
- the Ω-coefficient form of WP2 Task 2.2;
- the direct φ-evaluation from its defining property $\mathrm{ev} \circ (\varphi \otimes \mathrm{id}) = \mathrm{ev}$, with full associator bookkeeping (`docs/honest_twist_check.py`).

Agreement of four independent assemblies on sixteen data sets is the current standard of evidence for a formula, and it is what caught the defective transcription of Ω recorded in the plan. It is *not* a proof, and the proposal does not treat it as one: the kernel proof of WP2 Task 2.3 is what settles the formula, and the scripts are demoted to regression tests.

A second, independent technique is the Temperley–Lieb check: in $\mathrm{TL}(\varphi)$ one can compute the loop scalar of Fibonacci with no F-data at all, and it gives $\kappa_\tau = \varphi$ against the naive φ^2. It certifies the snake normalisation and nothing else, which is exactly the scope claimed for it.

6.6 Parallel adversarial exploration

The project uses a workbench in which several agents attack one problem in independent directions, register their direction on a shared board, and submit reports that a coordinating agent accepts only after re-deriving or kernel-checking the claim. Two things are worth recording, because both bear on the proposed programme.

What it bought. The pointed-case value of the twist, $J = \delta(\kappa) = \kappa_a\kappa_b/\kappa_c$, was settled in this mode by three independent assemblies; the identification of the trace identity with (M) emerged from it; and four independent evaluations of the J-diagram agreed, which is what isolated the defective Ω transcription.

What it got wrong. The same mode produced claims that did not survive checking: that $X^{**} \cong X$ canonically in any rigid monoidal category (false — this is precisely pivotality, and adopting it would have made the signature vacuous, as it had already done once); a scalar formula for ψ that returns 1 on channels where the honest twist is -1; and the assertion $J \equiv 1$ for pointed categories, which is false and was replaced by $\delta(\kappa)$. Each was caught by the acceptance rule, not by the generating agents.

Consequence for the programme. The technique is used for exploration and for independent assembly, never as evidence. Every claim in §4 marked "kernel" has been re-proved or re-checked by the coordinating agent; claims marked "script" are labelled as such; and no report is merged on the strength of its own text. The proposal's WP3 and WP4 explicitly budget for this: parallel exploration of the positivity step and of the cover, with kernel acceptance.

6.7 Statement fidelity as a method

The four defects are instances of four general failure modes, and naming them is part of the proposed contribution.

F1. Vacuity. The hypothesis is inconsistent, so the theorem is true and empty. Detected by inhabitation, not by proof.

F2. Drift. The formal statement is a different, usually stronger, statement (a one-line axiom where the source has a coherence condition). Detected by proving the source's examples fail, or by checking that the statement is not provable without its hypothesis.

F3. Unrepresentability. The signature cannot express the general case (no label transport for $i^{**} \neq i$; no multiplicity). Detected by trying to state the general theorem, not by proving the special one.

F4. Normalisation. The statement is right but the gauge is wrong: a reciprocal where a matrix inverse is required, a sign convention that excludes the examples. Detected by the examples, and only by the examples.

The audit protocol that follows is mechanical and is written up in Appendix B. It has three gates: *inhabitation* (does the structure have a witness?), *sharpness* (is the hypothesis used, i.e. does dropping it admit a counterexample?), and *correspondence* (is the statement the source's, with its numbering and its quantifiers?). The third gate is the one a proof assistant cannot automate, and the one that caught D3.

6.8 Engineering discipline

- **Pinned toolchain.** Lean `v4.33.1`, Mathlib `0df444a3`, in both packages; no bump without updating both lockfiles.
- **No `sorry`, no `axiom`**, axiom budget `[propext, Classical.choice, Quot.sound]` enforced by `AxiomAudit.lean`, which is a build gate and not a report.
- **Per-file builds** with generous timeouts (an `import Mathlib` costs 20–60 seconds; the example files cost minutes), `weakLeanArgs = ["-j2","-s8192"]` to stay inside CI memory limits.
- **No module-private declarations**: the project avoids the bare `module` header, which makes declarations private and breaks downstream imports in a way that is hard to diagnose.
- **Commit messages as a proof log**, including retractions. The repository's history carries three commits whose subject is a correction of the previous commit's mathematics; that is a feature, and the plan keeps the record rather than rewriting it.

7 Risks and mitigations

The programme is unusual in that its risk is concentrated in one step (WP3) and its deliverables are mostly independent of whether that step succeeds. The table makes the dependencies explicit.

	Risk	Mitigation / fallback	Level
R1	The positivity step of WP3 needs analysis that the skeleton cannot supply, and (K) is not reachable.	The conditional theorem `eno823_of_pseudoUnitary` already exists; the exact statement route isolates the gap as one scalar equation; if it stalls, D5 and D6 are published as "ENO 8.23 modulo positivity", which is still a new formal result.	med/high
R2	The signature change (general σ-twist, triangle, gauge fix) breaks the instance proofs.	Two-layer signature: the general one with σ, the strict one as its $\sigma = \mathrm{id}$ instance, with a transport lemma; instances proved once against the strict layer.	low

	Risk	Mitigation / fallback	Level
R3	Multiplicity > 1 expands to a second library.	Declared stretch goal; pursued through the categorical bridge (WP5) instead of a scalar signature.	low
R4	Mathlib churn: #42150/#42191 merge with different names, or stall.	Vendor, keep byte-compatible, switch by replacing one import; no unmerged API in the core. Contribute the spherical layer if it stalls.	low
R5	Elaboration cost: `decide` over 3^n label tuples grows fast; example files already cost 100 s.	Exact arithmetic in $\mathbb{Z}[\sqrt{d}]$; per-file builds; abstraction barriers so the core does not import the examples; `native_decide` forbidden.	low
R6	Parallel exploration produces confident false claims (it has: three).	Acceptance protocol: kernel re-proof or independent recomputation before merge; regression test per claim; claims labelled kernel/script/conjecture in the blueprint.	low
R7	The PF chain (#39914–#39922) is not mergeable in the project's window, so `FPData` stays a hypothesis.	Keep `FPData` as an explicit hypothesis with a per-example certificate; the theorem statements are already phrased to take it as an argument.	low
R8	Single-investigator scope over 24 months.	Priority order D1–D6 (core theorem) before D7–D12 (additive); WP4 and WP5 are parallelisable and separable.	med

The stop condition. If by month 16 the T-matrix domination of WP3 Task 3.2 has not been formalised, the programme pivots to WP5-first: build the categorical bridge, state ENO 8.23 for PivotalCategory, and leave the analytic step as a named hypothesis with the Yang–Lee sharpness theorem. That outcome is publishable and useful; the proposal does not treat it as failure, but it must be decided at month 16 rather than at month 23.

8 Timeline

Twenty-four months, eight quarters. Bars mark active work; the load is deliberately uneven, with WP2 and WP3 overlapping by design (the derivation of J is what makes WP3's statements well-posed, and WP3's bound is what tests WP2's formula).

Work package	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
WP0 Foundations	■	■						
WP1 Signature	■	■	■					
WP2 Derived twist		■	■	■	■			
WP3 Analytic core			■	■	■	■	■	
WP4 Cover & grading				■	■	■		
WP5 Bridge & upstream					■	■	■	■
WP6 Instances & tests	■	■	■	■	■	■	■	■
WP7 Dissemination				■		■	■	■
Milestones			M1, M2	M3	M4	M5	M6	M7

9 Expected deliverables

Twelve deliverables, each with a month and an acceptance criterion. The first six constitute the core; the rest are additive and independently valuable.

	Deliverable	Acceptance criterion	Month
D1	**Core library v1.0**: fusion rings; Perron–Frobenius dimension with `FPData` discharged; skeletal systems; pivotal and spherical layers; the twist reduction; ENO 8.21.	Clean build; no `sorry`; 100% of audited declarations at `[propext, Classical.choice, Quot.sound]`.	4–20
D2	**Reproducibility kit**: pinned toolchain, CI gates, clean-tree script, per-file harness, archived artifact with a DOI.	A fresh clone builds and audits without manual steps.	4
D3	**Signature v2**: triangle axiom; general double-dual permutation σ; general twist pentagon; gauge-invariant duality normalisation.	All six existing instances re-verified; the three new axioms proved for each.	8
D4	**Derived twist**: the comparison ψ, the ratio formula, the closed form Ω, the four laws and the general pentagon.	No hypothesis on J remains; the derived J agrees with four independent evaluations on sixteen data sets.	12
D5	**ENO 8.21 unconditional**: T-matrix domination, the bound $\lvert d_i\rvert \le f_i$ from the square root a, the sum identity, the equality case.	Proved with no `DimEigen` hypothesis.	16
D6	**ENO 8.23, exact form**: existence, sphericity, positivity, dim = FPdim, uniqueness.	`ENO823Exact` proved; `YangLee.not_pseudoUnitary` and `YangLee.not_eno823` as sharpness witnesses.	20
D7	**Cover and torsor**: the pivotal cover is a fusion category with a canonical pivotal structure and doubled dimension; pivotal structures form a torsor under $\mathrm{Hom}(U_{\mathcal{C}}, \mathbb{k}^\times)$; the splitting criterion.	The cover is a `FusionSystem` with a `PivotalStructure`; the criterion is a theorem, not a `def`.	18
D8	**Mathlib contributions**: the pivotal/spherical bridge; Perron–Frobenius dimension; the vendored patch deleted.	`MathlibPatches/` contains only imports.	24
D9	**Instance and regression suite**: sixteen data sets, including the two script-level witnesses of §4.5 formalised, and a multi-multiplicity datum.	Census harness green; one regression test per defect D1–D4.	18
D10	**Papers**: (a) the formalisation of ENO 8.23; (b) a negative-results note ("ENO 8.23 without pseudo-unitarity is false", with the kernel-checked Yang–Lee and $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$ witnesses); (c) a methodology note on statement fidelity.	Submitted: (b) by month 12, (a) by month 21, (c) by month 24.	12–24

	Deliverable	Acceptance criterion	Month
D11	**Blueprint and documentation**: statement-to-source table, module dependency graph, gauge conventions, a worked tutorial on adding an instance.	A reader can add a new instance without reading the core.	12
D12	**Open-problem dossier**: formal statements of ENO Conjecture 2.8 and its variants, with the known partial results (2-torsion criterion, cover) and the constraints any counterexample must satisfy.	Reviewed against the literature; every claim labelled proved/checked/conjectured.	24

What is not a deliverable. A proof of ENO Conjecture 2.8; the weak Hopf algebra input (ENO §4–5, Theorem 2.10); the general multiplicity case; and the classification of rank-three pivotal categories beyond the inequality used as a test. These are out of scope and are stated as such in the blueprint, so that a reader cannot mistake the boundary of the result.

10 Significance

For the classification programme. Proposition 8.23 is used as a black box in a large and growing literature; a kernel-checked proof would let downstream work cite a machine-checked theorem instead of a two-page argument whose normalisations are easy to get wrong. The present project's own history is the evidence: four defects, two of them in axioms, one of them turning the target into a tautology. A reader who trusts the literature's normalisations is trusting something that a careful formaliser, working from the source, got wrong four times.

For the mathematics. The formalisation has already produced statements the literature leaves implicit: that (K), the per-channel identity, is *not* invariant under the vertex gauge and survives only the duality normalisation $\lambda_{i^*} = \lambda_i$; that (M), the summed identity, is strictly weaker than the target and is satisfied by Yang–Lee; that the trace identity of ENO's proof is exactly (M); and that the correct general twist pentagon carries the associator on both sides, with the F-free form being a self-dual-label special case. None of these is a new theorem of fusion categories in the sense of a new mathematical result, but each is a precision about hypotheses that the informal literature elides, and precision of this kind is what makes a black box safe to use.

For Mathlib and the formalisation community. The categorical layer is arriving now (#42145, #42150, #42191, #42192). What is missing after it lands is exactly what this proposal builds: the finite-data layer, the examples, the counterexamples, and one substantial theorem that exercises the layer. The Perron–Frobenius material and the spherical layer are independent upstream contributions; the sixteen verified data sets are the regression suite that the categorical definitions will need.

For the author's programme. The sibling Haagerup–Izumi development formalises the analytic side of a classification theorem (coefficient identities, reduction to finitely many algebraic equations). This proposal formalises the algebraic side of the same story (dimensions, pivotal structures, sphericity). Together they cover the two halves of a verified pipeline from fusion rules to a categorical structure, and the Haagerup–Izumi categories are the natural large-scale instance for WP6.

Broader impact. Verified anyon data — F and R symbols with a checked pentagon and checked pivotal structure — is the input to proposals for topological quantum computation, and the failure mode there is precisely an unchecked normalisation. The library does not build a quantum computer, but it does make the statement "these symbols satisfy the axioms" a machine-checked one, and that is the part which is currently done by hand.

Limitations, stated plainly. The theorem is formalised for multiplicity-free data in a skeleton; the general case is a stretch goal through the categorical bridge. The analytic input of ENO is formalised, but its weak Hopf algebra prerequisite is taken as a hypothesis. And the work is a formalisation: its novelty is in precision, verification, and counterexample, not in a new route through the mathematics.

Acknowledgements

The preliminary results reported here were developed in the `fusion-category-lean` repository, whose `PLAN.md` records both the results and the retractions; the retractions are part of the evidence and are kept. The Haagerup–Izumi development in the sibling package is the work on which the feasibility argument rests.

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A Module inventory and dependency graph

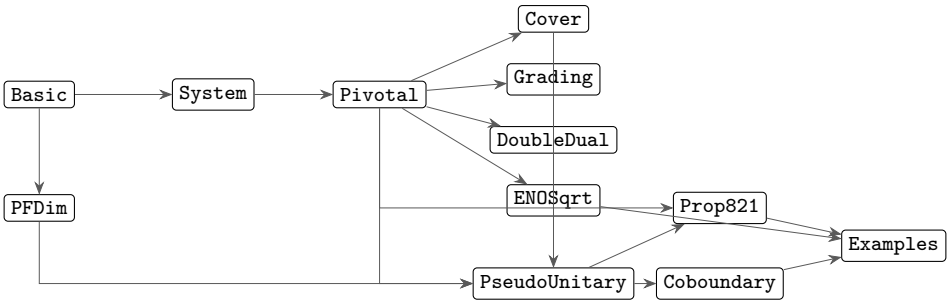


Figure 2: The import spine of the library (transitive imports suppressed; `Sanity`, `Step4Checks` and the individual `Examples/` files are leaves off this spine). `Examples/` is the umbrella over Fibonacci, Ising, Ostrik, $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$, Fibonacci–Ising and the step-4 checks; Yang–Lee and the pseudo-unitary instances are imported separately by the audit.

Module	Lines	Decls	Content and status
`Basic`	229	26	fusion ring, fusion/FP matrices, `IsPFEigenvector`, `FPData`, global dimension. Complete; `FPData` is a hypothesis.
`System`	451	19	`FusionSystem`: F, `Finv`, snakes, J; $\mathrm{Vec}_{\mathbb{Z}/2}$ inhabitant with the 512-tuple pentagon. Complete.
`Pivotal`	191	21	`PivotalStructure`, `dim`, `IsSpherical`, `isSpherical_iff`, `pVal_dual`. Complete.
`PFDim`	49	5	`PFUnique`, φ and its uniqueness. Complete (conditional on `FPData`).
`Cover`	128	8	`S4Data`, `CoverObj`, forgetful functor, `coverGlobalDim_double`. Scaffolding: no `PivotalStructure` on the cover yet.
`Grading`	153	8	universal grading, `pVal_dual`, two `def`s recording open targets. Formerly three axioms; now axiom-free.
`DoubleDual`	155	9	the older derived twist in the strict self-dual normalisation. Lags the ψ-derivation; to be rewritten against `Finv`.
`Coboundary`	331	15	`twistNorm`, `dim_eq_twistNorm`, `eno823_of_twistNorm` and its converse; the pointed `KappaMultiplicative` layer retained for reference. Complete.
`Prop821`	474	21	cyclic symmetry, `fpdim_char`, the weighted-norm bound, the equality case, `tMatrix`, `mult_iff_twist`. Complete modulo `DimEigen`.
`PseudoUnitary`	452	37	`ENO823`, `ENO823Exact`, `IsPseudoUnitary`, `IsPseudoUnitaryPivotal`, `eno823_of_pseudoUnitary`, `pivotal_ext`. Complete as a conditional reduction.
`ENOSqrt`	232	13	`SqrtData`, b, `Tmat`, d, `b_sq`, `trace_iff`, `toPivotal`. Complete; existence of a is assumed.
`Sanity`, `Step4Checks`	136, 147	8	numerical and consistency checks. Complete; imported by the umbrella.
`Examples/FibonacciIsing`	951	115	Fibonacci and Ising over $\mathbb{R}$: every field proved, spherical, $\dim = \mathrm{FPdim}$. Complete; $\sim$100 s.
`Examples/YangLee`	419	39	the refutation of the un-hypothesised target. Complete.
`Examples/VecZ2Twisted`	86	11	two pivotal structures, $\dim g = \pm 1$. Complete.
`Examples/Ostrik`	154	21	the rank-three ring $K(1,0,1)$ and its `FPData`. Complete.
`Examples/`$\cdots$	638	38	Fibonacci, Ising, pseudo-unitary instances, $Z/3$ note.
`MathlibPatches`	55	0	stubs for #42150 and the PF chain. Stub: WP0/WP5.
Total	**5,231**	**414**	23 files; build 3,032 jobs; audit 73 declarations.

B The audit protocol

The protocol is a sequence of gates applied to every declaration before it is merged. Its purpose is to catch the failure modes F1–F4 of §6.7, none of which is caught by a successful build.

Gate 1 — Inhabitation. Every structure introduced gets a witness in the same commit, and the witness is a case from the literature or a standard construction, not a degenerate one. For `FusionSystem` the witness is $\mathrm{Vec}_{\mathbb{Z}/2}$; for `PivotalStructure` it is $p \equiv 1$ on a datum with $J \equiv 1$ and a datum without. A structure with no witness is not merged.

Gate 2 — Sharpness. Every hypothesis is either used essentially or is recorded as unused. A hypothesis H is *used essentially* if dropping it makes the statement false on a witness. The two witnesses carried for this purpose are Yang–Lee (dropping pseudo-unitarity) and $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ (dropping multiplicativity). Where no counterexample is known, the declaration says so.

Gate 3 — Correspondence. Each theorem is entered in the statement-to-source table with (i) the source, (ii) the source's numbering in the edition used, (iii) the quantifiers and the carrier, (iv) any hypothesis of the source that is absent and why. The ENO numbering trap (8.23 in the Annals, 8.22 on the arXiv) is recorded there. This is the gate a proof assistant cannot automate, and the only one that catches drift.

Gate 4 — Axiom audit. `AxiomAudit.lean` prints the axioms of every exported theorem. The build gate is:

```
cd FusionCategory
lake build
lake env lean AxiomAudit.lean > /tmp/audit.txt 2>&1
grep -v "propext, Classical.choice, Quot.sound" /tmp/audit.txt
```

A non-zero exit from the final `grep` usually means *no matches* — i.e. success — and must not be reported as a failure of Lean. Two failure modes are worth recording: a `#print axioms` applied to an `axiom` declaration itself errors, and a name whose module is not imported returns `Unknown constant`; both have been mistaken for real findings in this project.

Gate 5 — Build hygiene. No `sorry`, no `axiom`, no bare `module` header, warnings permitted. The full build must pass from a clean tree (WP0 Task 0.2), not merely incrementally.

Gate 6 — Regression. Each defect of Table 1 has a named test that fails if the defect returns: `VecZ2.system` (D1), the 512-tuple pentagon (D2), `VecZ2Twisted.system` (D3), `swapped_Finv_comp_ne_one` (D4).

C The main statements

For reference, the target statements in mathematical form, with their Lean names.

Pseudo-unitarity. `IsPseudoUnitary S h` is the scalar equation

$$\sum_i \kappa_i \, \kappa_{i^*} \;=\; \sum_i f_i^2,$$

with no pivotal structure in it.

The bare target. `ENO823 S h` is

$$\exists\, P : \text{PivotalStructure } S, \quad P.\text{IsSpherical} \ \wedge\ \forall i,\ \dim_P(i) = f_i.$$

The exact target. `ENO823Exact S h` states: for every pseudo-unitary S there is a unique pivotal structure P with

$$P.\text{IsSpherical}, \qquad \forall i,\ \dim_P i > 0, \qquad \forall i,\ \dim_P i = f_i.$$

The conditional form already proved. `IsPseudoUnitaryPivotal P` is the conjunction

$$(\forall i,\ \dim_P i > 0)\ \wedge\ \Big(\forall i, k,\ \sum_j N_{ij}^k \dim_P j = \dim_P i \cdot \dim_P k\Big),$$

and `eno823_of_pseudoUnitary` proves that these two conditions on one pivotal dimension, together with `PFUnique`, give `ENO823`.

ENO's input. `SqrtData` carries $a\colon \mathrm{Fin}\, n \to \Bbbk^\times$ with $a_i^2 = g_i$ and the monoidality of a^2 stated against J^2; then b, `Tmat` and d are

$$b_c^{ij} = J_c^{ij}\frac{a_i a_j}{a_c}, \qquad T_c^{ij} = N_c^{ij} b_c^{ij}, \qquad d_i = a_i \operatorname{coev}_i \operatorname{ev}_{i^*},$$

and `trace_iff` states that $d_i d_j = \sum_c T_c^{ij} d_c$ for all i, j is equivalent to (M) in the duality normalisation $\kappa_i' = \operatorname{coev}_i \operatorname{ev}_{i^*}$.

The derived twist. The object WP2 must construct is

$$J_{ab}^c \ =\ \frac{\psi_{a,b}|_c}{\psi_{b^*,a^*}|_{c^*}}, \qquad \psi_{a,b}\colon (a\otimes b)^* \to b^*\otimes a^*,$$

with ψ the unique evaluation-intertwining isomorphism; the closed form in terms of F and the restricted inverse block M_b^{-1} is WP2 Task 2.2.

An ASCII rendering of the headline declaration. The Unicode binders are spelled out; this is a rendering for readability, not the source.

```
-- the target, in the form the library will carry it
structure ENO823Exact (S : FusionSystem n k) (hFP : S.ring.FPData) : Prop where
  pivotal    : IsPseudoUnitary S hFP ->
                 exists P : PivotalStructure S, P.IsSpherical
  positivity : forall (P : PivotalStructure S), P.IsSpherical ->
                 (forall i, dim P i = hFP.fpdim i) -> forall i, 0 < dim P i
  unique     : forall P Q : PivotalStructure S,
                 (forall i, dim P i = hFP.fpdim i) ->
                 (forall i, dim Q i = hFP.fpdim i) -> P = Q
```